University of Cape Town
Department of Computer Science

Computer Science CSC3003S
Class Test 1, 2025

Answer all questions.
Approximate marks per question are shown in brackets.

This test has 6 pages.

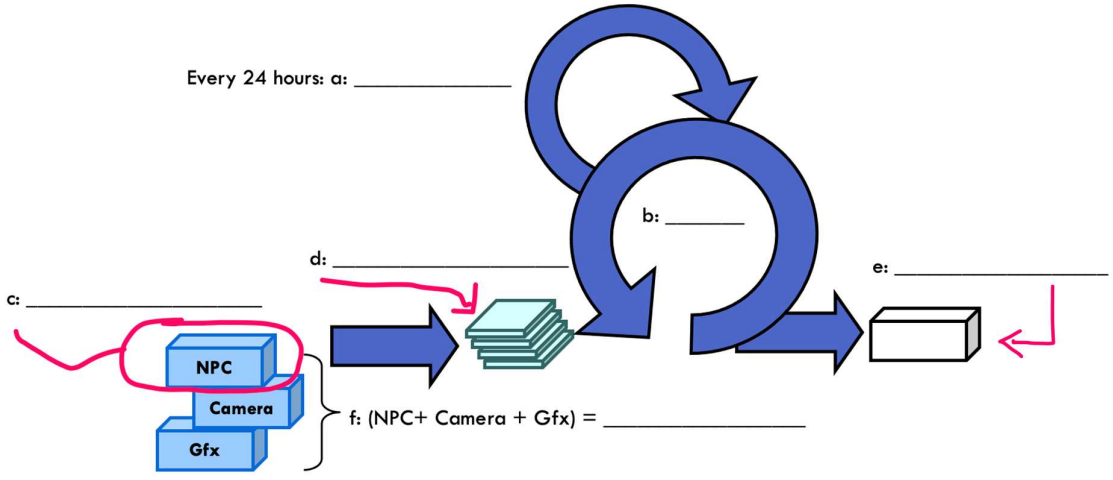
Time: 45 minutes **Marks: 29**

Question 1 [8]

a. Explain the difference between a "process" and a "project" [2]

b. In his book, The Mythical Man Month, Brooks explains how adding more developers to an overdue project delays it even more, why is this? [2]

c. Consider the below diagram of the Scrum methodology. [4]
Add four out of six missing labels, the scribbles point to the part to be named.
Write down at least 4 out of 6 labels, out of (a) to (f), in your answer book.
Make sure to write the letter next to your label.



Refer to the text below and answer the questions relating to it:

As discussed in the lectures, MCAS (Manoeuvring Characteristics Augmentation System) is a software system on the Boeing 737 Max family of planes that can alter the pitch (nose up/down) of the plane to make it behave and feel exactly like the older Boeing 737 NG planes. By making the new 737 Max feel the same as the old 737 NG plane, Boeing customers did not have to train their pilots for the new Max, but could have their pilots use their existing training for older 737 types instead, thus saving customers money.

However as designed the MCAS system was too powerful, and after MCAS caused two plane crashes, the FAA calculated that if not redesigned the MCAS system would have crashed 15 planes over its lifetime and killed over 2000 passengers. In reality, it crashed 2 planes in a 5 month timespan and killed 346 people. From a financial point of view, Boeing suffered ~$ 20 Billion in direct costs and ~$ 60 Billion in lost sales (which may – or may not – be recouped in the future).

Question 2 [7]

a. *You might very well think that Boeing's MCAS software development team did not perform a risk analysis prior to deploying their final MCAS prototype in the 737 Max planes sold to customers.*
Devise a risk strategy and based on that, advise Boeing management on what actions to take to manage the risk associated with this deployment. [2]

b. *The MCAS software development team spend considerable effort designing the MCAS features such that neither the aviation oversight board (the FAA), nor the pilots flying the 737 Max planes were aware of its existence and its true power.*
In so doing, which ISO 25010-2011 quality aspect did the developers ignore? Name the quality aspect and explain how its improves software. [2]

c. Explain what a "*wicked problem*" is. Are the dual crashes of Lion Air and Ethiopian Airlines caused by the MCAS system due to a "*wicked problem*" Explain why or why not. [3]

Question 3 [3]

What is aspect oriented programming (AOP) and how does it solve crosscutting concerns? You do not get points for merely mentioning the words: 'pointcut', 'advice' and 'aspect', instead explain how it works. [3]

Question 4 [4]

Consider the following lambda:

```
final var a = new ObjectX();
var my_lambda = (int i)-> {
    a.doStuff(i);
};
```

a. Write a class equivalent to this lambda in either Java or C++ (whichever one you are most comfortable with; note: the syntax does not have to match 100%), but do include all needed method(s), data member(s) and all needed visibility specifier(s). [4]

Code for question 5 on the next page.

```
[...] //includes
class Pear {
  int weight;
public:
  Pear(const int weight): weight(weight) {}
  int juice() const { return weight / 2; }
  void takeJuice(const int amount) { weight -= (amount * 2); }
};
class Grapes {
  int count;
public:
  Grapes(const int count): count(count) {}
  int wine() const { return count * 2; }
  void takeWine(const int amount) { count -= (amount / 2); }
};

using Fruit = std::variant<Pear, Grapes>;

class Print {
public:
  void operator()(const Pear& p) const { //worker method
    std::println("Hmm juicy pear {}", p.juice());
  }
  void operator()(const Grapes& g) const {
    std::println("{} grapes hold {} wine", g.count(), g.wine());
  }
};
class Squeeze {
    const int amount;
public:
    Squeeze(const int amount): amount(amount) {}
    void operator()(Pear& p) const { p.takeJuice(amount); }
    void operator()(Grapes& g) const { g.takeWine(amount); }
};

void PrintAll(const std::vector<Fruit>& fruits) {
    for (auto& f: fruits) { std::applyPattern( Print(), f); }
}

void SqueezeAll(std::vector<Fruit>& fruits, const int amount) {
    for (auto& f: fruits) { std::applyPattern( Squeeze(amount), f);}
}

int main() {
    std::vector<Fruit> fruits;
    fruits.push_back( Pear{10} );
    fruits.push_back( Grapes{20} );
    fruits.push_back( Grapes{30} );
    PrintAll(fruits);
    SqueezeAll(fruits, 1);
    PrintAll(fruits);
}
```

Question 5 [7]

Consider the code on the previous page.

a. *The std::applyPattern is not the real name of the method used in this pattern.* Which OOP design pattern does this code model? Explain what problem this pattern solves. [3]

In the above code the `std::variant` is a polymorphic class. As discussed in the lecture it consists of a tag field (in the example tag = tag_circle or tag_square) and space to accommodate one (and only one) of the classes it can model (in the example either `Square` or `Circle`).

b. Name the proper CS term for `std::variant`? (do not use the word 'variant') [1]

Notice in the above code how none of the classes use (or need) inheritance. In this variant (pun intended) of the well-known XXX design pattern this aids 4 of the 5 S.O.L.I.D. principles.

c. For 3 out of those 4 principles, list its full name and explain how that particular SOLID principle is satisfied by the structure of the design pattern in the example code on the previous page. [3]

Question 6 [6]

In the lectures we've discussed data oriented design (DOD aka DOP).

Redesign the following game pseudo-code such that it adheres to the principles of data oriented design. The syntax does not have to be 100% correct.
If the example code shows '...' your redesigned code should also show ...
Briefly explain the changes you made (how do they improve the code DOD-wise?)
The below code is Java code.

```
//imports
public class Sprite {
  public Integer x, y; //x,y is middle of the sprite
  public Integer h, w; //height, weight
  Bitmap picture; //no animation yet, just movement
  void Move(int speed, int direction) {...}
  boolean Overlaps(Sprite s) {...}
}
abstract class baseZombie extends Sprite {
  public Integer speed;
  public Integer direction; //1..360 degrees
  abstract void display();
  abstract Integer Overlaps(Sprite s); //returns damage on hit
}
//...
many different zombie types: class ZombieX extends baseZombie {...}
//...
public class Player extends Sprite {
  final Integer speed = 100; //fixed speed
  public Integer direction; //1..360 degrees
  public Integer health = 100;
}

public class Game {
  final int maxZombieCount = 10000; //never more than 10'000 zombies
  Sprite player;
  ArrayList<baseZombie> zombies;
  Game() {...}
  void step() {
    player.Move(...);
    for (var z: zombies) { z.Move(...); }
  }
  void doPlayerHit() {
    var oldHealth = player.health;
    for (var z: zombies) { player.health -= z.Overlaps(player); }
    if (player.health != oldHealth) { ... }
  }
  void display() { //runs 30x per second
    doPlayerHit();
    player.display();
    for (var z: zombies) { z.display(); }
    if (player.health <= 0) { gameOver(); }
  }
}
```